Criterion A – Planning

Staff dining hall routine system

Define the problem:

The client, Mr. Weatherby is a member of the current dining hall supervision system at the school, is inefficient, relying on assistant house parents to manually allocate slots to tutors based on their availability. **(Refers to Appendix A – Transcript of Initial Interaction with the Client)** This process, although thorough, is prone to issues such as unfilled slots and uneven workload distribution, especially for unpopular time slots like early mornings and weekend dinners. **(Refers to Appendix A – Transcript of Initial Interaction with the Client)** The primary goal is to develop a software solution that automates this allocation, reducing the administrative burden and ensuring a fairer, more efficient scheduling process.

Currently, the system involves assistant house parents gathering availability information from tutors and manually assigning them to five daily supervision slots from Monday to Saturday. While this allows tutors to indicate their preferences, the manual nature of the process makes it difficult to prevent conflicts, such as double bookings or missed assignments, and often results in some staff members being overburdened. **(Refers to Appendix A – Transcript of Initial Interaction with the Client)** Popular slots fill up quickly, leaving less desirable slots difficult to allocate, which further complicates the process and affects the overall efficiency.

The solution lies in creating an automated software system that streamlines the scheduling process. This software would collect and store tutors' availability, automatically allocate them to appropriate slots, and ensure a balanced distribution of duties among all tutors. **(Refers to Appendix A – Transcript of Initial Interaction with the Client)** By implementing such a system, the allocation process would become more efficient and transparent, reducing the workload on assistant house parents and minimizing scheduling conflicts. Furthermore, the software could include features like conflict alerts and real-time adjustments, making it easier to manage any last-minute changes and ensuring all dining hall supervision slots are consistently covered. **(Refers to Appendix A – Transcript of Initial Interaction with the Client)**

Rationale for the proposed solution:

intended to use java for this solution. It should handle large amounts of data, for example the timetable of teachers and information for each slot of data. Java enables Object Oriented Programming, which

allows encapsulation, polymorphism and inheritance. It is not only effective and able to handle large amounts of data and do complex calculations to fit all the slots but also enables User interfaces.

I propose that the implementation of this software can make a beneficial contribution to the system as it addresses systems based on reliable software **(Refers to Appendix A – Transcript of Initial Interaction with the Client).**

Word count*: 425 words*

Success Criteria: **(Refers to Appendix B – Approval of Successful criteria)**

1. The system must ensure that all dining hall supervision slots are filled for each day (Monday to Saturday), covering breakfast, lunch, and dinner.
2. Workloads must be distributed fairly among all tutors, ensuring no tutor is assigned more than four or five supervision slots per week.
3. Tutors must specify their unavailability, and the system must prevent scheduling during these times, avoiding any conflicts where multiple tutors are unavailable for the same slot.
4. The system must avoid scheduling tutors during other commitments (teaching or recurring pastoral duties) and handle fixed slots for specific tutors with weekly commitments.
5. It should allow manual intervention for unfilled slots. **(Refers to Appendix B – Approval of Successful criteria)**
6. The system should distribute unpopular slots (e.g., early breakfast, Saturday dinner) fairly across tutors to share the burden.
7. A user-friendly interface should allow for simple input of availability and automatic, transparent slot allocation, with feedback when conflicts arise.
8. Tutors' availability data, provided by the assistant house parents, will serve as the basis for scheduling decisions.
9. The system must enable comparisons between manual and automated allocations to assist in measuring efficiency.
10. A mechanism for random assignment of tutors to slots should be implemented while respecting availability and fairness constraints.
11. Tutor preferences for specific time slots should be considered to accommodate personal or logistical needs.
12. Tutors could be represented by a unique identifier to simplify input and allocation without revealing personal data in the system.
13. The system should generate reports or schedules that can be shared with tutors and assistant house parents, clearly displaying assigned slots and unfilled gaps.